# Overview

Separate proteins by size (SDS-PAGE) → transfer to PVDF → probe with antibodies → image by chemiluminescence.

**Mechanism:** Sodium dodecyl sulfate (SDS) denatures proteins and gives them uniform negative charge, so migration depends only on molecular weight.
**Optimized for PTPRB** — large, heavily glycosylated, low abundance. Settings marked ⚠️ are PTPRB-specific; use standard values for normal targets.

**Estimated timeline:**

| Stage | Time |
|---|---|
| Sample preparation (lysis) | ~30–45 min |
| Gel electrophoresis | ~1–1.25 h |
| Transfer | ⚠️ 1.5 h (standard: 30–45 min) |
| Blocking | 1 h |
| Primary antibody | 1 h (RT) or overnight (4°C) |
| Washes + secondary antibody | ~1.5–2 h |
| Detection/imaging | ~15 min - 1h |

**Equipment needed (not consumed):** SDS-PAGE gel tank and power supply, transfer apparatus (wet), PCR machine or heat block, tabletop centrifuge, sonicator, rocking platform, chemiluminescence imager,

**Safety notes:**

- BME (2-mercaptoethanol) is a flammable, foul-smelling reducing agent — handle in a fume hood and keep in the flammable cabinet.
- Methanol (used for PVDF activation and in transfer buffer) is toxic and flammable — use in a ventilated area.
- SDS and acrylamide gels are irritants.

# Materials and Preparation

| Reagent | Cat# | Storage |
|---|---|---|
| Cell lysis buffer | # 9803S | Aliquoted in small tubes, -20 storage room |
| Protease inhibitor | # PPC1010-1ML | -20°C, aliquoted |
| TGX gel | #4561095 | 4°C fridge B, storage room, on the door |
| Laemmli sample buffer | #1610737 | DNA gel bench shelf |
| 2-mercaptoethanol (BME) | # M3148-25mL | Flammable cabinet |
| Protein ladder (up to 250 kDa) | #1610374 ladder | -20°C fridge B storage room |
| 10x Running buffer | #1610732 | DNA gel bench shelf (double check cat# since there are similar buffer) |
| PVDF membrane 0.2µm for normal proteins | #1212781 (⚠ **0.45 µm**) # T13087 | DNA gel bench shelf #1620174 needs methanol socking up |
| Filter paper | #88600 | DNA gel bench shelf |
| TBST | # 9997S (or prepare, see below) | Stock without tween20 on my shelf |
| 10 x Transfer Buffer | #1610734 | DNA gel bench shelf |
| SuperSignal West Femto | #34094 | 4°C; Zhang lab is more gentle |
| milk powder | #30620074 | chemical shelf |

**Antibody needed based on experiment**

| Antibody | Cat# | Dilution | Storage |
|---|---|---|---|
| Anti-PTPRB (rabbit) | PA568309 | 1:1000 | –20 °C |
| Anti-rabbit-HRP | 32460 | 1:2000 ⚠ (standard 1:5000) | 4 °C |
| Anti-GAPDH (rat) | Zhang lab aliquot | 1:10000 | 4 °C box |
| Goat anti-rat | Zhang lab aliquot | 1:5000 | 4 °C box |

**1x TBST (100 mL):** 10mL 10x TSBT (# 9997S) + 90mL water
20 mM Tris, 150 mM NaCl, 0.05–**0.1%** Tween 20

**Self made 10× Transfer Buffer** (500 mL) :15 g Tris base + 72 g glycine + $H_2O$ to 500 mL. Do not pH (~8.3). RT or 4 °C, several months.

**1× Transfer Buffer (2 L)**: 200 mL 10x Transfer Buffer + 400 mL methanol + 1400 mL $H_2O$. Make during the gel run.

**1x running buffer (1 L):** 900 mL water + 100 mL 10x running buffer (#1610732)

**Blocking buffer** (5% milk, 50mL): 2.5g milk + 5mL 10xTBST + 45mL water

# 1. Sample Preparation

mcf7_protocols > Cell Lysis

1. Dilute lysis buffer: 900 µL dd$H_2O$ + 100 µL 10x (9803S). Add 10 µL protease inhibitor **fresh**.
2. Resuspend 10 M cells in 0.4 mL buffer. 10 min at 4 °C rocking → sonicate in short pulses, **on ice**.
3. Spin 14,000 g, 5 min, 4 °C. Keep supernatant on ice.
4. Quantify by Bradford/BCA.
   *10/24/26: 500k MCF7 + 20 µL lysis buffer → ~3.5 mg/mL*
5. Store aliquots at –80 °C. Avoid freeze-thaw. Thaw **quickly** — prolonged time on ice shifts pH and degrades band quality.
   **Troubleshooting:** Weak signal downstream usually starts here — under-lysis, slow thawing, or bad quantification.

# 2. Gel Loading and Electrophoresis

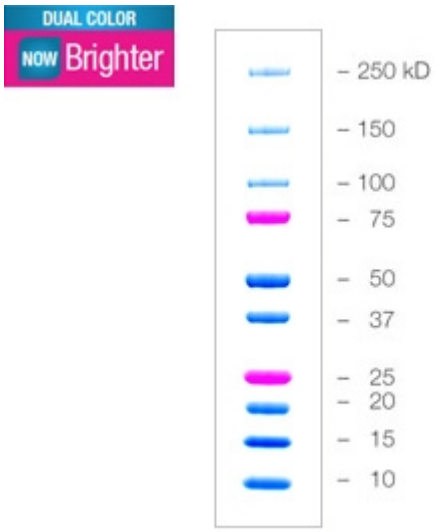
> **Load volume:** 20 µL total per sample. Standard 30–50 µg protein/lane.
> ⚠️ PTPRB: ~10 mg/mL in the 20 µL loading sample (200 µg cel lysate). Run GAPDH from the same aliquot in a separate lane.

1. Prepare samples : mix Laemmli buffer and BME first and then add to your sample.
   - 10 µL 2x Laemmli + 0.5 µL BME + 10 µL sample, **or**
   - 5 µL 4x Laemmli + 0.5 µL BME + 15 µL sample
   - Ladder: 5 µL 4x Laemmli + 0.5 µL BME + 5 µL $H_2O$ + 10 µL ladder (1610374)
2. Denature samples at 95°C for 5 min in a PCR machine. No need for protein ladder to denature
3. Assemble the gel cassette (TGX gel, #4561095, 4–20%): remove comb and green tape → two cassettes, **short side inward** → fill inner chamber with 1x running buffer, then outer to the 2-gel line.
   *Buffer dam（plastic plate） works instead of a second gel, but it hides bubbles.*
4. Load **10 µL** of the 20 µL. Load protein ladder first, then **every other lane** (2, 4, 6…).
   (I used to load with 15µL, but there is always signal in the adjacent lane likely due to sample leakage)

5. Run 124 V voltage constant (~72 mA), ~1 h, until dye front nears bottom.
   ⚠️ Full 1 h for PTPRB — more separation in the high-MW region.
   - Check for bubbles after powering on.

- While the gel runs, prepare **2 L** of transfer buffer (see recipe above 1x transfer buffer)

Protein ladder (# 1610374)




## 3. Transfer to Membrane

**Materials:** SDS-PAGE gel, transfer apparatus, wash buffer (TBST), membrane — nitrocellulose or PVDF, methanol (for PVDF), filter paper (#88600).
Youtube: gel transfer or backup

3.1. Remove gel from cassette. Equilibrate sponges, filter paper, membrane, gel in **separate** bins of transfer buffer. Fill trays with ice.
*Immuno-Blot membrane (1620174): methanol 15–30 s first, then transfer buffer.*

3.2. Build the sandwich
**white (+) sponge → filter paper → membrane → gel → filter paper → sponge → black (–)**
Membrane sits on the positive side of the gel — protein runs out of the gel toward it. then close the cassette

3.3 Gel tank on ice → insert cassette, **black side to black electrode** → add ice pack inside tank.

3.4 Run 100 V / 200 mA constant. *In practice: holds 200 mA, voltage settles ~85 V.*
⚠️ **1.5 h for PTPRB** (standard 30–45 min). Keep cold the whole time.

Figure 1: assembly

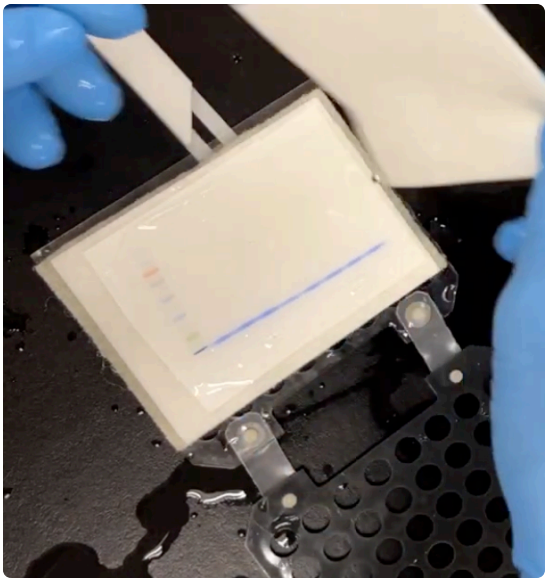
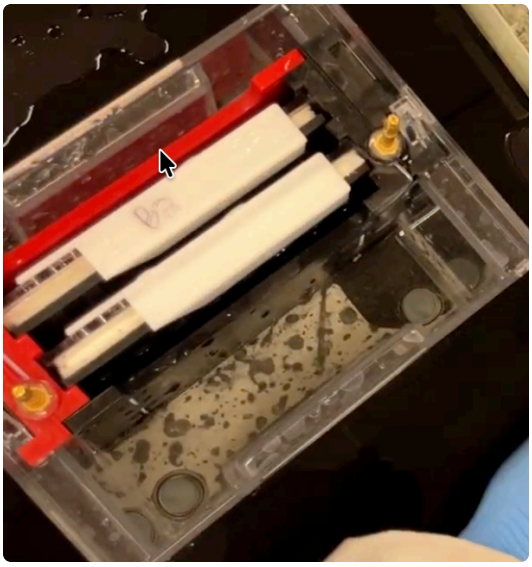


Figure2:Gel tank



Note:

> - PTPRB is large and heavily glycosylated protein, which moved out of the gel more slowly than typical mid-size protein
> - Keep the apparatus cold (ice block or cold room) throughout, since heat buildup is more of a concern over this longer run.
> - No protein on membrane → check sandwich orientation, current, and time.

⚠ **Waste disposal: Transfer buffer should be collected in chemical waste container and request pick up**

## 4. Antibody Staining and Detection

**Goal:** specifically label the target protein and visualize it.

- 5mL will cover the surface of membrane in western blot container

4.0 Prepare
50mL **blocking buffer** (2.5g milk + 5mL 10xTBST + 45mL water)
50mL 1x TBST

4.1. **Block** 1 h RT, gentle rocking
Milk blocks better, but check the datasheet — phospho-specific antibodies fail in milk.

4.2. **Primary antibody** in blocking buffer: overnight 4 °C or 1h RT
overnight 4 °C gives better signal:noise, preferred for PTPRB
PTPRB antibody (host: rabbit) -> 1:1000
anti-GAPDH antibody -> 1:10000

4.4. Wash the membrane 3x with TBST, 5 min each, with rocking

4.5. **Secondary antibody** (species-matched HRP) in blocking buffer, 1 h RT.
Goat anti-Rabbit IgG (H+L) HRP secondary -> 1:2000 (standard is 1:5000)
anti-rat secondary antibody -> 1:5000

4.6 Wash the membrane 3x with TBST, 5 min each, with rocking

4.7. **Imaging:** mix West Femto (34094) 1:1 fresh, apply evenly, 1–5 min RT, protect from light.

- How ECL works: HRP oxidizes the luminol in the presence of peroxide, producing light as a byproduct. This light is captured by your chemiluminescence imager
- Classic ECL is enough for GAPDH. Femto is for low-abundance targets and saturates fast.

4.8. **Image.** Start short (1s), increase as needed.

- image at Brain lab Urey Hall, **or**
- image at zhang lab: Zhang lab ECL reagent gives weaker signal
  - Chemiluminescence mode → start 10 s → cancel once bands appear → check Gallery (top left)
  - Colorimetric → manual, 4 s → brightfield
  - Gallery → select both → **Merge**
  - Process in Image Lab (free)

**Troubleshooting:**

| Problem | Fix |
| --- | --- |
| Weak/no signal | Revisit lysis: under-lysis, degradation, bad quantification. Load more. Longer primary incubation. |
| No protein on membrane | Sandwich orientation, current, transfer time |
| High background | More/longer washes, Tween to 0.1%, dilute antibody further |
| Nonspecific bands | Dilute antibody, extend blocking, verify antibody specificity |
| Smeared lanes | Overloading or incomplete denaturation |
| Smiley front | Voltage too high or run too long |
| Signal in adjacent lane | Load 10 µL not 15 µL; skip lanes |